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PAPER

Classification of attention deficit/hyperactivity disorder based on EEG signals using a EEG-Transformer model*

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E-mail: mlnmxmg@163.com, mtong@ent.cuhk.edu.hk and sx.chen@siat.ac.cn**Keywords:** attention deficit/hyperactivity disorder (ADHD), electroencephalogram (EEG), transformer, attention mechanism**Abstract**

Objective. Attention-deficit/hyperactivity disorder (ADHD) is the most common neurodevelopmental disorder in adolescents that can seriously impair a person's attention function, cognitive processes, and learning ability. Currently, clinicians primarily diagnose patients based on the subjective assessments of the Diagnostic and Statistical Manual of Mental Disorders-5, which can lead to delayed diagnosis of ADHD and even misdiagnosis due to low diagnostic efficiency and lack of well-trained diagnostic experts. Deep learning of electroencephalogram (EEG) signals recorded from ADHD patients could provide an objective and accurate method to assist physicians in clinical diagnosis. **Approach.** This paper proposes the EEG-Transformer deep learning model, which is based on the attention mechanism in the traditional Transformer model, and can perform feature extraction and signal classification processing for the characteristics of EEG signals. A comprehensive comparison was made between the proposed transformer model and three existing convolutional neural network models. **Main results.** The results showed that the proposed EEG-Transformer model achieved an average accuracy of 95.85% and an average AUC value of 0.9926 with the fastest convergence speed, outperforming the other three models. The function and relationship of each module of the model are studied by ablation experiments. The model with optimal performance was identified by the optimization experiment. **Significance.** The EEG-Transformer model proposed in this paper can be used as an auxiliary tool for clinical diagnosis of ADHD, and at the same time provides a basic model for transferable learning in the field of EEG signal classification.

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1. Introduction

Attention Deficit/Hyperactivity Disorder (ADHD) is a neurodevelopmental disorder, the most common behavioral disorder and the second most common chronic disorder in children and adolescents, that manifested primarily by incongruence with developmental levels [1]. Depending on the symptoms, a subtype of ADHD that does not involve impulsivity and hyperactivity is further classified as Attention Deficit Disorder (ADD) [2]. According to the survey, the global prevalence of ADHD is about 5% and most are found at school age, among that 65% of the patients will continue to adulthood, which seriously impairs the individual's attention function, cognitive process, and working memory [3–5]. Thus, it has a continuous negative impact on the study and life of patients. Only an accurate diagnosis can further treat the patient and gradually eliminate the effects of the disease. At present, in clinical diagnosis, physicians mainly make diagnoses based on the fifth edition of the 'Diagnostic and Statistical Manual of Mental Disorders' (DSM-5) formulated by the American Psychiatric Association [6]. Typically, a physician's clinical assessment of ADHD takes at least an hour, and a global shortage of trained specialists leads to significant delays in diagnosis and thus disease delays [7, 8]. Studies have shown that ADHD can be diagnosed by electroencephalogram (EEG) and magnetic resonance imaging (MRI), which can assist physicians in clinical diagnosis and further improve the efficiency of diagnosis [9–13].

Diagnosis using machine learning (ML) technology requires manual extraction of feature data of the input information, selection of important features according to different standards, and finally using the features for ML classifiers for automatic EEG diagnosis [14–16]. However, ML methods are difficult to cope with the processing of high-dimensional data, so researchers have proposed deep learning (DL) algorithms for processing high-dimensional data. DL technology can automatically extract the features of input information and learn useful information for classification [17–19]. A growing number of studies also use deep learning models for EEG classification [20, 21]. EEG signal data is different from two-dimensional image data, it contains a lot of spatial and temporal information. Convolutional neural network (CNN) can extract spatial information with translation invariance through convolution, but its feedforward network structure does not consider the correlation between data, and the network output is only related to the current input. Therefore, CNN cannot effectively process temporal information [22, 23]. There are recurrent connections in the recurrent neural network (RNN), which makes the network have memory and can capture time information. However, the RNN model is complex and cannot perform parallel computing, and there are problems such

as gradient disappearance and gradient explosion [24].

The Transformer deep learning model was first applied to natural language processing (NLP) in 2018, and achieved excellent performance in the field of computer vision, including image classification, segmentation detection, image super-resolution, etc [25–27]. The key to the superior performance of the Transformer model is that the model abandons the traditional RNN and CNN structures, and directly selects the attention mechanism. The attention mechanism can perform parallel computing, which will greatly improve computing efficiency and reduce the time cost of the training process. At the same time, compared with the traditional feature extractor, the attention mechanism can calculate the correlation between any two words and solve the long-term dependence problem. Since EEG signals contain a lot of time information, Transformer can be applied to the field of EEG signals.

At present, the research based on the Transformer model in the field of EEG has been applied to EEG emotion recognition and EEG classification. Liu *et al* proposed four Transformer transformation models for EEG emotion recognition, among which the spatial-temporal attention model achieved an optimal classification accuracy of more than 90% on DEAP and SEED data [28]. Song *et al* built a Transformer-based EEG decoding S3T model, and achieved a classification accuracy of over 84% on both sets of public data sets [29]. Xie *et al* built five Transformer-based deep learning models, and achieved the best accuracy rates of 83.31%, 74.44% and 64.22% in the two-class, three-class and four-class tasks, respectively [30]. Delvigne *et al* processed the spatiotemporal information of EEG signals based on the Transformer model and achieved a classification accuracy of more than 74% in the comparison of commonly used models [31].

The main contributions of this paper are as follows:

- (1) This paper proposes an EEG-Transformer EEG signal classification model, uses the attention mechanism to extract EEG signal features, and makes full use of EEG spatiotemporal information to achieve high-performance classification;
- (2) Through the ablation experiment, the internal structure of the EEG-Transformer model was deeply analyzed, and the functions of the internal modules and the impact on the performance of the model were mastered, making the model structure clearer and more transparent;
- (3) In the process of optimizing the performance of the EEG-Transformer model, the model achieves the best classification performance through the adjustment of the model structure, which

provides a theoretical basis for model adjustment and modification of other types of EEG data and can be used as a basic model in the field of EEG classification.

2. Methods

2.1. The dataset

Since deep learning requires a large amount of EEG data from patients, we chose to use an open-source dataset from the Technical University of Dresden in Germany. The data structure of all data can be viewed at <https://osf.io/6594x/>. The acquired data were approved by the Ethics Committee of the Technical University of Dresden, and informed consent was obtained from all participants and their guardians in accordance with the Declaration of Helsinki.

The entire study sample included 144 children who were diagnosed and classified by psychiatrists and psychologists based on standard clinical indicators, home and school interviews, questionnaires, and tests of IQ and attention. Among them, there were 48 children with ADHD, 52 children with ADD, and the remaining 44 were healthy control (HC) children. There were no differences in the age, IQ or sex distribution of the children in the groups. And the patient has no other severe or acute psychiatric comorbidities (such as autism, convulsions, depressive episodes, etc.)

The original data were preprocessed by independent component analysis to remove noise and were divided into 33 902 trials, of which 10 742 trials were included in the ADHD group, 13 031 trials were included in the ADD group, and 10 129 trials were included in the control group (HC group). The acquisition time of each experiment was 1.5 s, the number of electrode channels was 56, and the sampling rate was 256 Hz. The format of each group of data is arranged in the form of two-dimensional channel number and sampling point.

This study adopted the method of five-fold cross-validation to realize the division and processing of data. There are 33 902 experiments in total, 6000 of which were divided into test sets. The remaining 27 902 trials will be divided equally into five parts, four of which will be the training set and one will be the validation set. After five training epochs, the validation set goes through all the training data.

2.2. Structure of the transformer model

Deep learning models in the field of NLP are mainly used for tasks such as machine translation, text generation, and information retrieval [32]. Since the input and output of the NLP model are both language sequences, the sequence-to-sequence (Seq2Seq) model architecture is usually used to process and analyze the data correspondence. The Transformer model also has a Seq2Seq structure,

which is mainly divided into two parts: encoder and decoder. The Encoder structure is used for data dimension reduction and feature extraction, and the Decoder structure is used to decode the Encoder feature information and output the target sequence according to the model [33]. This study proposes an EEG-Transformer based on the traditional Transformer model. The specific structure of the model is shown in figure 1. EEG-Transformer cancels the Decoder structure in the traditional model, and realizes the Transformer Block with the feature extraction function of EEG signals based on the Encoder structure; uses the global maximum pooling (GMP) layer to focus on the salient feature information and realize the dimensionality reduction operation. Then use the fully connected layer and Softmax classification layer to finally realize the overall EEG-Transformer model architecture. This section mainly introduces the working principle of the main modules of EEG-Transformer.

2.2.1. Position embedding layer

The EEG signal contains time-related information, and the feature extraction attention mechanism in the Transformer model will randomly extract various features of the data while ignoring the feature data of the location information. Therefore, to make the Transformer use the location information, position embedding is added before the data is input into the attention mechanism. In this study, a learnable positional encoding was used to add a trainable weight matrix of the same dimension to the input data, and the positional encoding parameters were continuously optimized during the model training process.

2.2.2. Multi-head attention layer

The Multi-Head Attention module uses the attention mechanism to realize the feature extraction of EEG data, and the implementation process is as follows. First, the input vector is represented as X , and X is multiplied by three different random weight matrices W_Q , W_K and W_V to obtain query vector Q , key vector K , and value vector V , where d represents the model dimension. The specific formulas of Q , K , and V were shown in (1)–(3):

$$Q = W_Q X \quad (1)$$

$$K = W_K X \quad (2)$$

$$V = W_V X. \quad (3)$$

After obtaining the Q , K , and V vectors, we calculated the dot product of the query vector Q and the key vector K , then divide by $\sqrt{d_k}$ for normalization. After that, we obtained the attention weight through the Softmax function and finally compared the weight with the value vector V . The multiplied value is the output of the Attention mechanism:

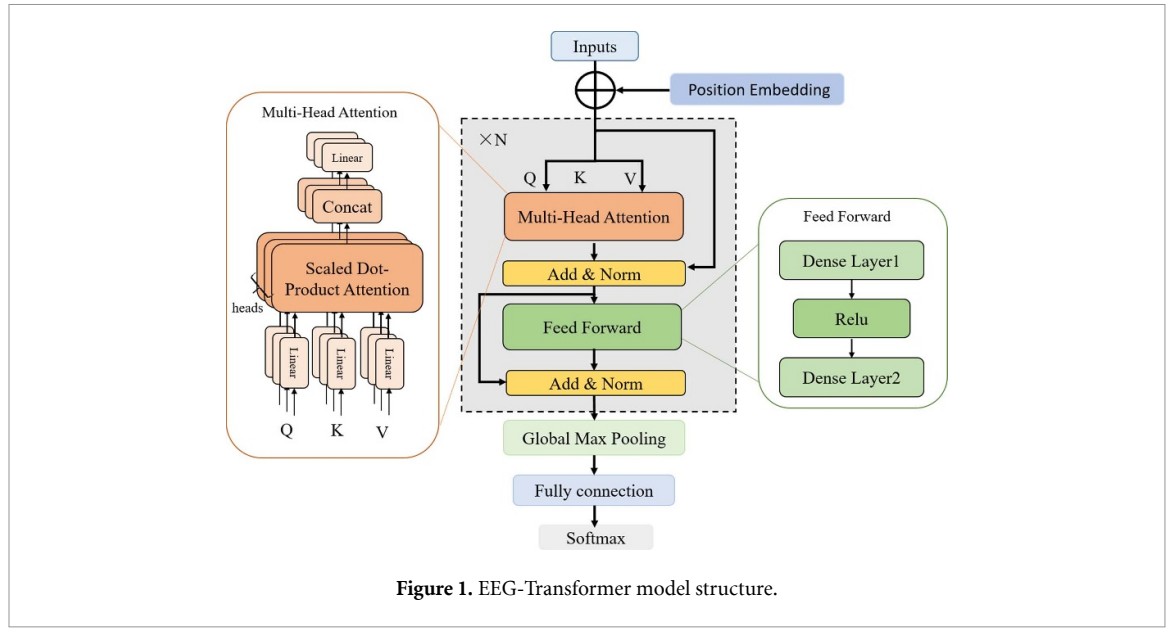


Figure 1. EEG-Transformer model structure.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V. \quad (4)$$

The attention modules in (4) were stacked in parallel as Multi-Head Attention modules, allowing the model to focus on information in different subspaces.

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O \quad (5)$$

$$\text{head}_i = \text{Attention}(Q, K, V). \quad (6)$$

Where head_i is the scaled dot product attention in (4), and W^O is the projection parameter matrix.

2.2.3. Feed forward layer (FFN)

The FFN was a two-layer fully connected layer, and the activation function of the first layer was ReLU. The corresponding formula was as follows (7):

$$\text{FFN}(x) = \max(0, xW_1 + b_1) W_2 + b_2. \quad (7)$$

2.2.4. Add & Norm layer

The Add & Norm Layer was used twice for classification in the model, referred to as Add&Norm1 and Add&Norm2 for short. Simply put, Add & Norm Layer was composed of residual network and Layer normalization. The specific formula was as follows:

$$\text{Add\&Norm1} = \text{LayerNorm}(X + \text{MultiHeadAttention}(X)) \quad (8)$$

$$\text{Add\&Norm2} = \text{LayerNorm}(X + \text{FeedForward}(X)). \quad (9)$$

2.2.5. Global max pooling

The global max pooling layer performs dimensionality reduction processing on the output data of the upper layer module as a whole while retaining the

feature information that has the greatest impact on classification and inputs the data after dimensionality reduction into the lower layer classification module.

2.3. The comparative CNN models

In this study, three commonly used deep learning models for EEG signal classification were used. EEGNet consists of three convolutional layers. The first layer of two-dimensional convolution is used to achieve frequency filtering, and the second layer of deep convolution is used to connect the output feature map of the first layer of convolution to learn spatial filtering of specific frequencies. The third layer of separable convolutions is a combination of Deep convolutions. The separable convolution learns the time summary of each feature map separately and then uses the point convolution to optimally mix the feature maps to finally achieve the classification effect [34].

ShallowConvNet has a large convolution kernel, mainly composed of two layers of convolution. The first two layers of convolution complete the functions of temporal convolution and spatial filtering respectively. With two layers of convolution, square nonlinearity, average pooling layer, and logarithmic activation function, ShallowConvNet can learn the time structure of frequency band power changes during training, which is helpful for classification [35].

DeepConvNet is structurally similar to ShallowConvNet but consists of four layers of convolutions. The first layer is divided into two layers, which is beneficial to handle a large amount of input channel data. The second layer of convolution will perform weighted spatial filtering on the output of the first layer, but there is no activation function added between the first layer and the second layer, so it can also be regarded as a layer of complex convolution. The last two layers of convolution are similar to

Table 1. The structure of the model parameter.

Block	Layer	Size	Activation	Parameters
Position embedding	Embedding	(None,385,56)		21 560
Transformer block x6	Multi-Head Attention	(None,385,56)	Relu	76 328
	Add & Norm	(None,385,56)		112
	Feed forward	(None,385,56)		7288
	Add & Norm	(None,385,56)		112
	Total	(None,385,56)		83 840
Pooling	Global max pooling	(None,56)		0
	Dropout	(None,56)		0
Full connection	Dense	(None,64)	Relu	3648
	Dropout	(None,64)		0
Softmax	Softmax	(None,3)	Softmax	175

the ShallowConvNet convolution. DeepConvNet can extract a wider range of features from data to meet the needs of multi-level classification tasks [35].

2.4. Model training process

The details of the model parameter structure was shown in table 1. There were 506 883 model training parameters in total. The hyperparameter settings of the training process were respectively, epoch = 300, batch size = 256, the learning rate optimizer chooses Adam, and the learning rate = 0.001. At the same time, to prevent the model from overfitting, functions such as model check point, Early Stopping, and Reduce learning rate were added during the training process. Among them, the Model Check Point function would record the optimal parameters of the training process, and the early stop function would monitor the value of the Loss function during the verification process. If the loss did not produce an optimization improvement within 30 epochs, the training was stopped directly, and the reduced learning rate function monitored the value of the loss function of the verification set at the same time. If the valid loss had not been optimized within 20 epochs, the function changed the value of the learning rate to half of its previous value.

Were used as the training sets and 5580 trials were used as

The experimental environment used in this experiment is windows 10 Enterprise, 4 NVIDIA TITAN Xp GPUs were used for training, each with 12GB video memory, the programming language was python3.8, and the deep learning model was built using Tensorflow-gpu2.50.

2.5. Performance metrics

In order to accurately analyze the performance of the classification model in this study, Accuracy, Precision, Recall, F1-score, ROC curve, and AUC value were calculated:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (10)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (11)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (12)$$

$$\text{F1score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{recall} + \text{Precision}} \quad (13)$$

$$\text{FPR} = \frac{FP}{TN + FP} \quad (14)$$

$$\text{TPR} = \frac{TP}{TP + FN} \quad (15)$$

where TP = true positives; TN = true negatives; FP = false positives; FN = false negatives.

This article used the receiver operating characteristic (ROC) curve and area under curve (AUC) value to solve the phenomenon that the unbalanced distribution of positive and negative samples in the test set leads to deviations in the results. False positive rate (FPR) indicates the probability that all negative samples are predicted to be positive samples, and true positive rate (TPR) indicates the probability that all positive samples are predicted to be correct. By changing the classification threshold, a series of TPR, and FPR points are obtained. Draw the TPR-FRP curve according to the threshold value from small to large, that is, the ROC curve. The AUC value is the area under the ROC curve fluctuating between 0–1. The higher the AUC value, the better the discriminative ability of the model.

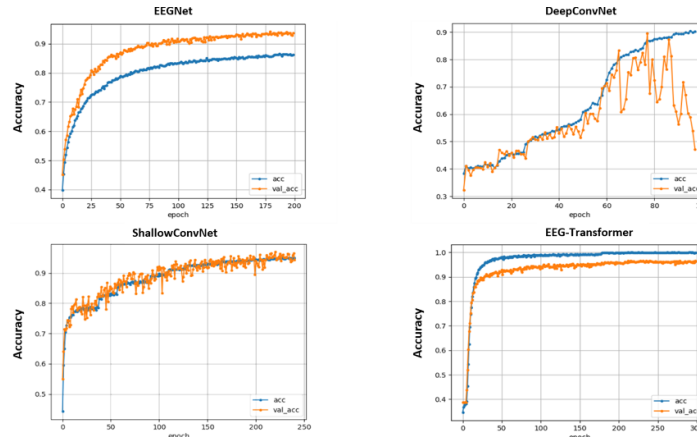
3. Experimental results

3.1. Performance comparison of different models

The parameters of the four types of models and the training time of a single epoch are shown in table 2. The model parameter for EEGNET, ShallowConvNet,

Table 2. Parameters and training time of the four types of models.

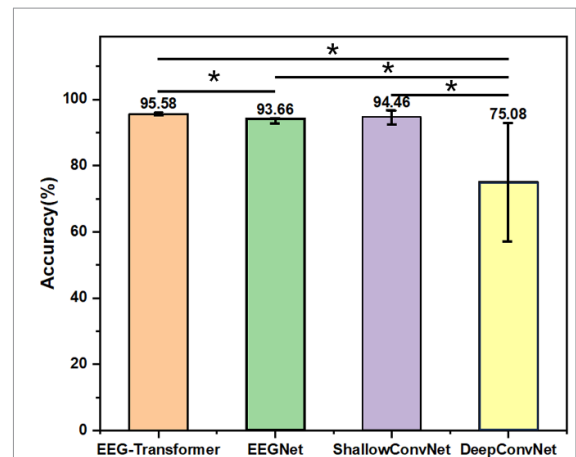
	EEGNet	ShallowConvNet	DeepConvNet	EEG-Transformer
Parameters	2659	23 923	180 278	528 423
Time(s)	6	8	9	30

**Figure 2.** The accuracy change curve of the training process of the four types of models.

DeepConvNet, and EEG-Transformer were 2659, 180 278, 23 923, and 528 423 respectively, while the single epoch training time for those were 6 s, 9 s, 8 s, and 30 s respectively.

The accuracy change curves of the four types of models during the training process were shown in figure 2. EEGNet converged and stabilized at about 100 epochs, and the accuracy of the verification set reached more than 90%, while the accuracy of the training set was significantly lower than that of the verification set. DeepConvNet continued to rise but the model did not converge. The classification of the validation set fluctuated greatly, and the influence of the final classification dropped to about 45%. The ShallowConvNet model had not yet converged, and the accuracy of the verification set was basically the same as that of the training set, but there were slightly unstable fluctuations. Compared with the aforementioned models, the EEG-Transformer model quickly converged within 30 epochs, the curve increased steadily and the verification set data was close to the training set data.

The classification accuracy of the four types of models was shown in figure 3. The accuracy rates of EEG-Transformer, EEGNet and ShallowConvNet were similar, $95.58\% \pm 0.52\%$, $93.66\% \pm 0.79\%$ and $94.46\% \pm 2.16\%$. The accuracy rate of DeepConvNet varied greatly, which was $75.08\% \pm 17.83\%$. At the same time, we used ANOVA to conduct quantitative statistical analysis on the four types of models. Among them, the P values between EEG-Transformer and EEGNet and DeepConvNet models were all less than 0.05, which proved that there were significant differences between these two groups of models. The p-value between EEG-Transformer and

**Figure 3.** Classification accuracy of the four types of models.

* indicates significantly different data

ShallowConvNet was greater than 0.05, and there was no significant difference between the two. The p-value between EEGNet and ShallowConvNet was greater than 0.05. According to the experimental results, EEG-Transformer was verified with the best performance.

The confusion matrix was shown in figure 4. The labels on the left side of the image represented the actual classification of the data, and the labels below represented the classification predicted by the model. Since the result was three classifications, the final confusion matrix is divided into 9 parts. Each part represented the accuracy rate when the actual value corresponds to the predicted value. The diagonal data represented the classification

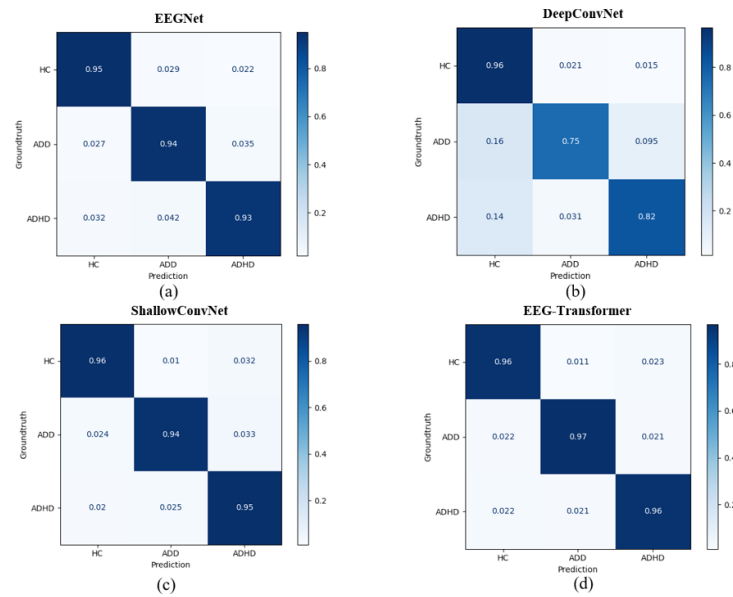


Figure 4. The confusion matrix of four types of models.

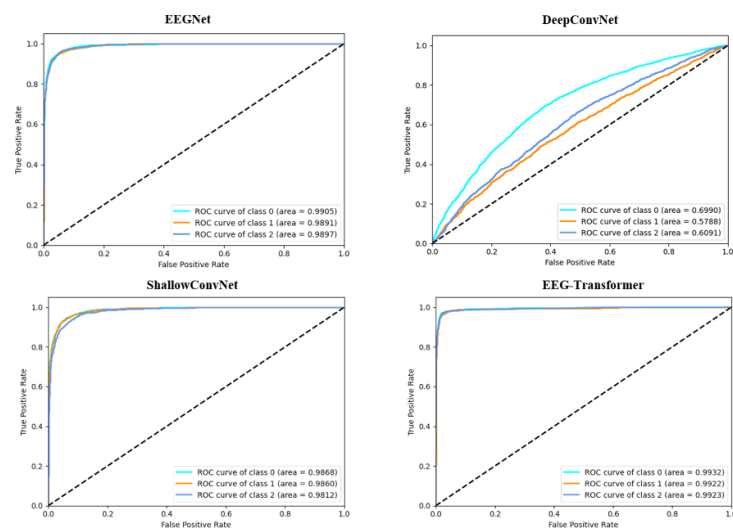


Figure 5. The results of ROC curve and AUC value. The blue line represented the HC group curve, the orange curve represented the ADD group, and the purple curve represented the ADHD group.

accuracy rate of the model. According to the specific accuracy data, DeepConvNet only achieved 75% and 82% classification accuracy for the ADD and ADHD groups, and the differences among the other three types of models were relatively small. Further comparison of Accuracy values showed that EEG-Transformer had the highest classification accuracy in the HC group, ADD group, and ADHD group (HC: 0.96; ADD: 0.97; ADHD: 0.96), with an average accuracy rate of 96.3%.

The ROC curve and AUC value further illustrated the classification performance of the model. As shown in figure 5, the curves of the four types of models were all higher than the dotted line in the figure, indicating that they were all effective. The DeepConvNet

curve was close to the midline, but away from the coordinate axis, and the average classification accuracy of the three groups reached 0.6290 (HC: 0.6990; ADD: 0.5788; ADHD: 0.6091). For the remaining three types of models, the EEG-Transformer's overall and local performance were the best indicators, with an average AUC value of 0.9926 (HC: 0.9932; ADD: 0.9922; ADHD: 0.9923).

3.2. Role of component modules in EEG-Transformer model

Ablation experiments are a commonly used control variable method in deep learning. Since the model is equivalent to a black box, it is difficult to analyze

Table 3. The results of the Ablation experiments. A check mark indicates a module that was kept from the experiment, and a cross indicates a module that was removed from the experiment.

Position embedding	Multi-Head attention	Feed forward	Add & Norm	GMP	GAP	Accuracy(%)
✓	✓	✓	✓	✓	×	95.58 ± 0.52
✓	✓	✓	✓	×	✓	84.08 ± 3.31
×	✓	✓	✓	✓	×	95.82 ± 0.71
✓	×	✓	✓	✓	×	39.12 ± 14.88
✓	✓	×	✓	✓	×	82.39 ± 32.47
✓	✓	✓	×	✓	×	16.23 ± 6.62

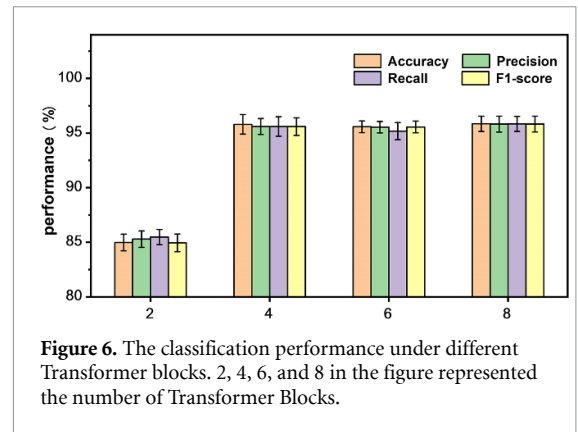
the performance of the module intuitively. For complex deep learning models, the impact of modules on overall model performance is studied by removing modules from the network [36–38]. This experiment studies the effect of choosing the GMP layer and the global average pooling (GAP) layer on the classification performance in the EEG-Transformer model, as well as the lack of Position embedding layer, Multi-Head attention layer, FFN, Add & Norm layers on model classification.

The results of the ablation experiment were shown in table 3. The results of the first and second rows can visually show the difference in the classification results using GMP and GAP. The classification accuracy rate using the GMP model was 95.58% ± 0.52%, while the classification accuracy rate using the GAP model decreased to 84.08 ± 3.31%. After deleting the Position embedding Layer, the accuracy rate did not decrease but slightly increased, and the accuracy rate was higher than that of the control group (Accuracy = 95.82% ± 0.71%). When the feed forward module was removed, the classification accuracy drops to 82.39% ± 32.47%. Relative to the decrease in precision, the standard deviation showed a more significant (±32.47%). When the Multi-Head Attention module was deleted, the overall classification accuracy of the model dropped significantly to 39.20%, with a standard deviation of 0.55%. Finally, after removing the Add & Norm module, the classification accuracy of the model was 38.38% ± 0.81%.

3.3. Optimization of transformer structure

In this section, we mainly study the effect of the number of Attention Heads and Transformer Blocks in the Multi-Head Attention Layer on the performance of the EEG-Transformer model. The experiment comprehensively measures the model through Accuracy, Precision, Recall, F1-score, model training parameters and single training time.

When studying the effect of the Transformer Block on the classification performance of the model, the Attention Head is set to 6 at this time, and the Transformer Block is tested in the order of sizes 2, 4, 6, and 8. The performance metrics in the experimental results are shown in figure 6, and the parameters and training time are shown

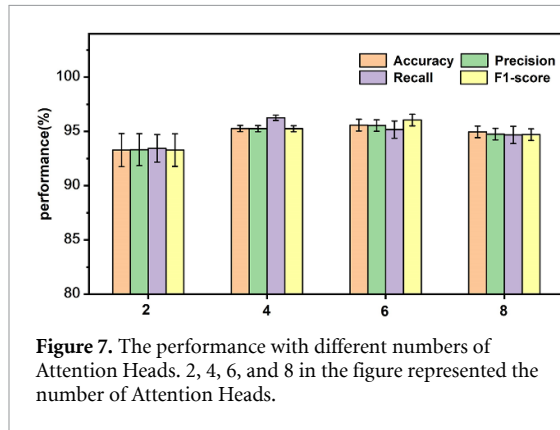


in table 4. When the Transformer Block is 2, the number of parameters is 171 523, the training time of a single epoch is 18 s, and the classification performance is low (Accuracy = 84.98% ± 0.76%, Precision = 85.29% ± 0.76%, Recall = 86.47% ± 0.70%, F1-score = 84.95% ± 0.81%). When the Transformer Block is 4, the number of parameters is 339 203, and the training time of a single epoch is 30 s, the performance of the model is significantly improved, and each index reaches about 95% (Accuracy = 95.80% ± 0.89%, Precision = 95.60% ± 0.72%, Recall = 95.60% ± 0.89%, F1-score = 95.59% ± 0.80%). The Transformer Block is increased to 6, the number of parameters is 506 883, and the training time of a single epoch is 45 s. At this time, the performance of the model tends to be stable, and the performance is slightly lower than that of the previous group (Accuracy = 95.58% ± 0.53%, Precision = 95.54% ± 0.53%, Recall = 95.17% ± 0.79%, F1-score = 95.56% ± 0.54%). When the Transformer Block is further increased to 8, the number of parameters is 674 563, and the training time of a single epoch is 58 s, the model performance is similar to the previous group (Accuracy = 95.85% ± 0.70%, Precision = 95.82% ± 0.73%, Recall = 95.81% ± 0.70%, F1-score = 95.83% ± 0.72%).

Set the Transformer Block to 6 layers, and experiment with the number of Attention Heads in the order of 2, 4, 6, and 8. The performance indicators of the four types of models are shown in figure 7,

Table 4. Different module numbers correspond to model training parameters (Param) and training time (T).

Number	2		4		6		8	
	Param	T(s)	Param	T(s)	Param	T(s)	Param	T(s)
Transformer Block	171 523	18	339 203	31	506 883	45	674 563	58
Attention Head	201 759	22	354 339	34	506 883	45	659 427	54



and the parameters and training time are shown in table 4. When the Attention Head is 2, the number of parameters is 20 179, the training time of a single epoch is 22 s, and the model performance indicators are all around 93% (Accuracy = $93.28\% \pm 1.53\%$, Precision = $93.32\% \pm 1.47\%$, Recall = $93.44\% \pm 1.27\%$, F1-score = $93.28\% \pm 1.51\%$). When the Attention Head is 4, the number of model parameters is 354 339, the training time of a single epoch is 31 s, and the model performance is further improved (Accuracy = $95.27 \pm 0.28\%$, Precision = $95.25\% \pm 0.28\%$, Recall = $95.26\% \pm 0.24\%$, F1-score = $95.25\% \pm 0.27\%$). When the Attention Head is 6, the number of parameters increases to 506 883, the training time of a single epoch is 45 s, and the model performance is slightly improved (Accuracy = $95.58\% \pm 0.54\%$, Precision = $95.54\% \pm 0.53\%$, Recall = $95.17\% \pm 0.79\%$, F1-score = $96.05\% \pm 0.54\%$). Finally, when the Attention Head increases to 8, the number of parameters reaches 659 427, and the training time of a single epoch is 54 s. Compared with the previous group, the model training time increased and the model performance decreased (Accuracy = $94.96\% \pm 2.42\%$, Precision = $94.75\% \pm 2.22\%$, Recall = $94.68\% \pm 2.31\%$, F1-score = $96.05\% \pm 2.27\%$).

4. Discussion

This paper constructs an EEG-transformer model for ADHD EEG signal classification tasks and compares it with the other three types of models through model evaluation indicators such as confusion matrix, the results verify that the EEG-Transformer model has the best classification performance. Through the ablation

experiment, the functions of the internal modules of the model were analyzed, the internal structure of the model was further understood, and the clear and complete model structure of the EEG-Transformer was demonstrated. Finally, in the process of continuous optimization of the model, the general law of model structure adjustment was discovered, which improved the transferability of the model.

4.1. Performance comparison of different models

In recent years, various effective deep learning models have emerged in the field of ADHD classification. Qin *et al* proposed a multimodal data fusion strategy and ensemble learning model (Trans3D-ensemble) to classify ADHD and normal, and achieved 74.5% accuracy on the ADHD-200 dataset [39]. Dubreuil-Vall *et al* constructed a CNN with a four-layer architecture that combines filtering and pooling, and the classification accuracy of the model reached $88\% \pm 1.12\%$ [21]. Riaz *et al* achieved a classification accuracy of 73.1% using the DeepFMRI deep learning model [40]. Saurabh *et al* proposed an improved deep learning-based Bidirectional Long Short-Term Memory model, which achieved a classification accuracy of 87.50% [41]. Vahid *et al* using the same dataset as in this paper, the study applied a deep learning model 'EEGNet' to distinguish ADHD subtypes from healthy controls with a classification accuracy of 83%. In the three-category task of distinguishing ADHD, ADD, and healthy controls, the accuracy was $69 \pm 28\%$ [42].

According to the results shown in table 2, the three types of models, EEGNet, ShallowConvNet and DeepConvNet, have low complexity and relatively small number of parameters, and the single epoch training time is below 10 s. The EEG-transformer model is complex, with the most parameters, and the training time is 30 s, which is almost three times that of other models. The other models have low complexity, so the training speed is faster. In order to improve the classification accuracy of the model, the EEG-Transformer chooses to increase the number of model parameters, which leads to an exponential increase in the training time.

In figure 2, we notice that the accuracy of EEGNet converged quickly, but the accuracy of the training set was lower than that of the validation set. According to the model structure analysis, it might be because there were too many regularized dropout layers in the training model. Dropout would randomly reduce 50% of

the upper model neurons during training and automatically turned off Dropout during verification so that the neural network could use all the neural networks leading to improved accuracy. The Accuracy curve of the DeepConvNet model did not converge, and the fluctuation of the verification set curve was much larger than that of the training set. The verification set proved that the classification performance of the model was low. Due to the complex structure of the DeepConvNet model, which consisted of four layers of convolution modules with different depths, the model was prone to overfitting when the amount of input data was relatively small, resulting in the accuracy of a validation set being much lower than that of the training set. The ShallowConvNet model was generally convergent, and the accuracy fluctuation of the verification set was small. Due to the simple structure of the model, there was no over-fitting phenomenon. It might be that the batch size setting 256 was too large, and it was difficult for the neural network to converge to a local optimum. EEG-Transformer could not only achieve fast convergence but also guarantee high classification accuracy and model classification performance. Among the four models, EEG-Transformer showed the best classification accuracy. Therefore, compared with the convolution-based neural network model, EEG-Transformer realized the parallel input of data in the process of extracting data features using the Attention mechanism.

Among the three types of CNN models, except for the poor classification result of DeepConvNet due to overfitting, the other two types of EEGNet and ShallowConvNet achieved higher accuracy. However, the classification accuracy of the EEG-Transformer in each group of data classification was the best. At the same time, according to the ROC curve and the corresponding AUC value, it could be further proved that the Attention mechanism in EEG-Transformer made full use of the time information contained in the EEG signal. Therefore, compared with the other three CNN-based models, EEG-Transformer had better classification performance.

4.2. Role of component modules in transformer model

The purpose of the ablation experiment is to explore the role of internal modules in the efficient classification of models, visualize the black box of the model, facilitate more specific and scientific changes and combinations of modules according to requirements, and maximize the advantages of the model.

The pooling layer can achieve data dimensionality reduction while retaining data information. The GMP layer selects the most important features, the GAP layer averages the data features, and the functions of the two are different. According to the experimental data, it can be concluded that the classification performance and classification stability of the model decreased after GAP treatment compared with GMP

treatment. Since GMP focuses on the most significant features in the data and focuses on the most critical fluctuation information of the signal, the classification characteristics of the signal can be grasped to the greatest extent, while GAP averages all the features, paying too much attention to the whole, but missing the information of the detailed function. Therefore, for brain electrical signals, it is more suitable to use GMP as a pooling layer.

In the traditional Transformer model, the Position embedding Layer can reflect the relationship between words and express the hidden connection between words through position coding. Therefore, an embedding Layer is added to EEG-Transformer to analyze the connection between EEG signals of different channels. After removing the Position embedding Layer, the accuracy of the model increased slightly by 0.3%, and the error increased by 0.2%, indicating that the embedding Layer did not significantly improve the classification performance of the model. The possible reason for the failure of Position embedding is that the model is complex and the number of parameters is large. After adding a weight matrix with the same dimension as the data, the model parameters increase slightly, and there is a slight overfitting phenomenon. Therefore, after removing the model, the classification performance of the model improves. However, after losing the location information of the data, the classification error also increases slightly, adding a certain amount of instability. In summary, although using the Position embedding Layer will cause a slight performance loss. For the sake of stability, this part of the module is retained in this model.

When the Multi-Head Attention module was removed, the performance of the model decreased significantly. Since the Multi-Head Attention module was the core module for Transformer to obtain EEG signal features, it was difficult for the model to extract the required feature information after removing it. From the perspective of feature extraction, Multi-Head Attention was an indispensable basic module in the entire model, which had a profound impact on the entire process of signal classification.

After removing the FFL, the overall classification accuracy of the model dropped by about 10% when compared with the basic control group. At the same time, the training process was unstable, and the classification results in the 5-fold cross-validation repeatedly showed that the classification performance was low. According to the classification results, it could be concluded that the training process of the model without Feed Forward was unstable, leading to the problem of gradient disappearance. Since the nonlinear ReLU activation function was added to Feed Forward, the model could be fitted nonlinearly, which was more suitable for the data. At the same time, it could solve the problem of gradient disappearance in the backpropagation process of optimizing model parameters. Therefore, the

Feed Forward module played a role in stabilizing the classification.

After removing the Add & Norm Layer, the classification accuracy of the model was only 38.38%, which was consistent with the effect of losing the Multi-Head Attention Layer and failing to maintain a high classification performance. The model used Add & Norm twice in the Transformer, which worked after the Multi-Head Attention module and Feed Forward respectively and adds the data before and after the Multi-Head Attention input for normalization. First, the residual network prevented network degradation and the problem of gradient disappearance/explosion. The normalization process of Layer normalization, on the one hand, could speed up the convergence of the model, and on the other hand, it could also stabilize the model and prevent the gradient from disappearing/exploding. Therefore, when the model lost the Add & Norm module, the normal gradient calculation could not be performed in the face of the complex EEG-Transformer network structure, which eventually led to poor performance. Hence, Add & Norm is essential for fast model convergence and improved stability.

4.3. Optimization of transformer structure

In the experiment of studying the influence of Transformer Block layers on the performance of the EEG-Transformer model, a comprehensive analysis was carried out through performance index, parameter quantity, and training time. When the Transformer Block is 2, although the training time of a single epoch is short, the performance index fluctuates around $85.42\% \pm 0.72\%$, and the model classification performance is low. When the Transformer Block is 4 layers, the overall performance of the model is higher than that of 6 layers, but in the process of comparing the error items, the error of the former is on average 0.2% higher than that of the latter, which proves that the model performance has a certain degree of instability. Finally, the classification performance of the model under the 8-layer Transformer Block is slightly higher than that of the 6-layer model, the average index is 0.4% higher, and the error is 0.12% higher. But the 8-layer Transformer Block model has too many parameters. It takes 24.2 h to complete 50 cross-validations training every 300 epochs, which consumes a lot of computer resources. With the continuous increase of modules, the complexity and training time of the model will also increase exponentially. Transformer Block is the main component of the model, including the Multi-Head Attention Layer, FFL and Add & Norm Layer. Every time the Transformer Block is added, it represents a uniform increase of its internal modules, so it has a great impact on the classification performance and model complexity, and the model classification effects of different Block layers vary greatly. Therefore, the adjustment of the Transformer

Block module is suitable for the EEG data training in the early stage of model training, and the general model framework suitable for the data is determined through different model layers.

In the experiment of fixing the Transformer Block and exploring the influence of the number of Attention Heads on the EEG-Transformer model, a comprehensive evaluation was carried out according to the model parameters, training time cost and classification accuracy. The larger the number of Attention Heads, the more complex the model, the more parameters, and the time cost of training will also increase. When the Attention Head is 2, the training time of a single epoch is only 22 s. When the Attention is increased to 8, the training time of a single epoch is 54 s. If the epoch is 300 times, it takes 4.5 h to complete a training session, while in 5-fold cross-validation, it takes 22.5 h to complete a complete training session. When the Attention Head is 2, the classification accuracy of the model reaches about 93%, and the model has preliminary classification performance. In the process of further increasing Attention Head, the classification performance of the models with Attention Head = 4 and Attention Head = 6 is similar, but the performance of the latter is relatively higher, which can achieve more accurate EEG classification. Finally, due to the complexity of the parameters of the 8 Attention Head models, overfitting will occur during the training process, resulting in a decrease in training accuracy and consuming a lot of computer resources and training time. Therefore, when the number of Transformer Blocks is constant, a small number of Attention Heads can complete the classification task, but when there are too many Attention Heads and the training parameters exceed the parameters corresponding to the amount of data, the loss of classification performance and time will result in cost. From the histogram in figure 6, it can be seen that the classification accuracy of the four groups of curves with different numbers of Attention Heads is similar, and all indicators are above 90%. The adjustment of the Attention Head is aimed at the classification details of the EEG-Transformer, which can improve the classification performance of the EEG-Transformer in a small range. Therefore, the Attention Head module is suitable for adjusting and optimizing the model details after the Transformer Block framework is determined. According to two sets of experiments, it can be further concluded that in the EEG-Transformer model architecture for ADHD data in this paper, the 6-layer Transformer Block and 6-layer Attention Head module can achieve high classification accuracy and high stability. At the same time, the model training time is moderate, which can make full use of computer resources.

The sample size of this study is 300, and a total of 33 902 sets of experimental data were obtained by dividing the data. Although the amount of experimental data has expanded, the smaller sample size has

not changed. The application of this data is feasible in the theoretical research stage, but if it is subsequently applied to the clinical diagnosis process, it is necessary to further increase the sample size, continuously train the model, and increase the visibility of the EEG-Transformer.

5. Conclusion

This study transferred the Transformer model from the field of NLP to the field of EEG classification in children with ADHD. At the same time, three types of EEG classification models based on CNN, namely EEGNet, DeepConvNet and ShallowConvNet, were compared in this paper. According to the comprehensive performance comparison of the Accuracy curve, confusion matrix and ROC curve during the training process, the EEG-Transformer model could quickly and stably realize high-performance EEG classification, which proved the effectiveness and parallelism and computationally efficient extraction of EEG time-series signal features of the attention mechanism in Transformer. Ablation experiments prove the necessity of using the Global Max Pooling Layer in EEG signal classification, and the Multi-Head Attention module and Add & Norm module play an important role in feature extraction and model stability in EEG signal classification. Finally, through the Attention Head and Transformer Block structure adjustment experiments, it can be concluded that when the EEG-Transformer model is applied to new data, it is first necessary to adjust the number of Transformer Block layers to preliminarily determine the overall framework structure of the model. Secondly, optimize the details of the model through the adjustment of the Attention Head, and determine the complete model through the classification performance and model complexity. For the ADHD data in this paper, the EEG-Transformer model performs best when the Transformer Block is 6 and the Attention Head is 6. The EEG-Transformer proposed in this paper can provide a reliable auxiliary method for clinical diagnosis, and at the same time provide a new idea for transfer learning in the field of EEG classification.

Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: <https://Osf.io/6594x/>.

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